

MULTIMODAL RECOGNITION

Week 4: Image Captioning

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MASTER IN
COMPUTER VISION
Barcelona



GitHub Repository

https://github.com/JordiVentosa/C5_Team3

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1. Introduction

1.1 Previous Week

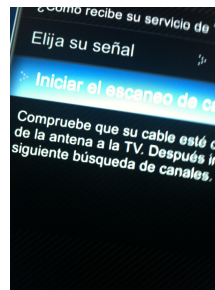
In the previous week, we started with the image captioning task using classic DL solutions. We adopted a progressive methodology to incrementally update our initial baseline.

This baseline originally consisted of a ResNet-18 encoder and a GRU decoder operating with character-level tokenization. Through systematic experimentation, we explored the impact of scaling the encoder to a ResNet-34, comparing GRU and LSTM decoder architectures, and modifying the tokenization level to evaluate subword and word representations. Furthermore, we analyzed the effect of introducing Bahdanau attention and conducted a final hyperparameter search.

Our results demonstrated that the most significant performance improvements came from structural changes, specifically the transition to higher token levels like words, rather than just hyperparameter tuning.

Ultimately, the experiments also highlighted a general tendency of these recurrent models to severely underfit the complexity of human language.

When compared to other groups, we realized we achieved bad results, for this reason we expect to significantly improve the results this week thanks to a better architecture and a renewed pipeline



Ini.Base: "A blrte"
Mod. Enc.: "A coite"
Mod. Dec.: "A alite s e a
oo o e isual
content."
Mod. Tkn.: "a screen screen a
a a recognize a"
Mod. Att.: "a computer screen
a a a a a"
Mod. Hyp.: "a computer
screen a a a a a"
GT: "A settings screen of
selecting a menu option in
Spanish text."



Ini.Base: "A blat o"
Mod. Enc.: "A ality issue ar o
eeee ni isaa onrnttt."
Mod. Dec.: "A alite s e a
oo o e isual
content."
Mod. Tkn.: "a bottle bottle a a
a a a a"
Mod. Att.: "a bottle of of a a
a a"
Mod. Hyp.: "a person with a a
a a a"
GT: "a black bottle with yellow
label placed on a fabric
materials"

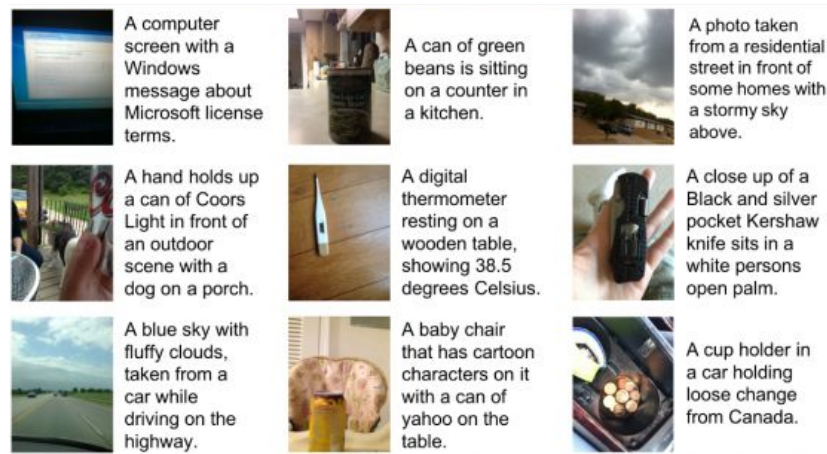
Previous Week Results				
Metric vs. Model	BLEU-1	BLEU-2	ROUGE-L	METEOR
Initial Baseline	0.059	0.043	0.177	0.114
Modified Encoder (ResNet34)	0.061	0.044	0.173	0.118
Modified Token Level (GRU Word)	0.201	0.136	0.286	0.196
Optimized Hyperparameters	0.201	0.135	0.279	0.196

1.1 Dataset Recall

As the previous week, we're using the **VizWiz dataset**. The dataset contains thousands of images with a ratio of **5 captions (or references) per image**.

The captions have an average length of **11.64 words** with a median at 10, with 99% of the captions contain **27 words** or fewer.

It's important to recall that the images of this dataset are taken by people with vision impairments, and therefore the images are usually distorsionated, blurry or undistinguishable in some extreme cases from noise. In general, are images taken without a consideration on good photography.



Split	Images	Captions	Origin
Train	21,087	105,435	90% train folder
Validation	2,344	11,720	10% train folder
Test	7,750	38,750	100% val folder
TOTAL	31,181	155,905	

2. Image Captioning using ViT-GPT2 architecture

2.1 ViT + GP2 inference

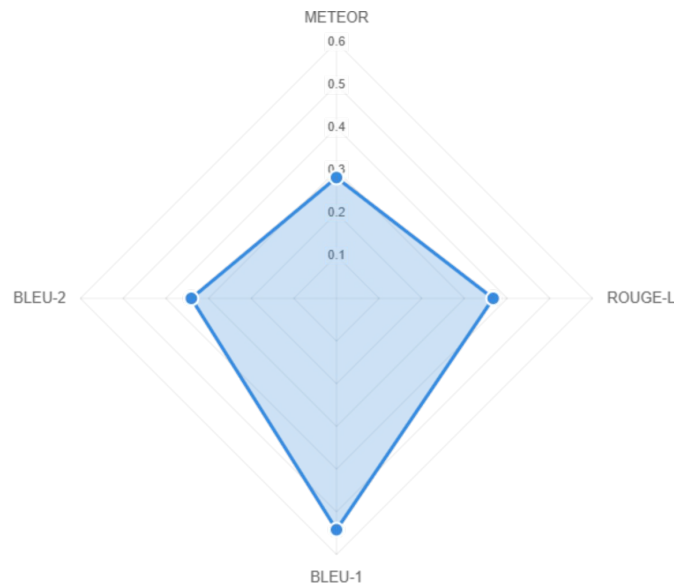
We started by implementing the ViT + GP2 image captioning model from github and testing the results on our validation set.

Initial results were pretty good already, especially BLEU1.

Looking just at the metrics, it is clear the model is recognizing things in the image, but it may not be properly describing them.

For example, a Bleu1 high score may indicate the model is recognizing general traits like colors, or certain objects, but because of the low meteor score it is not properly describing the scene, just naming objects or colors that appear.

The best way to understand what is going on is to check the actual captions to see what the model is doing.



	Meteor	Rouge-l	Bleu-1	Bleu-2
Baseline	0.283	0.367	0.542	0.340

2.1 ViT + GP2 inference

Checking the results, something becomes clear quite quickly, the model always try to predict what is in the image, but a lot of annotations actually say that the image is too bad to recognize anything.

This will probably be easily solved after fine tuning, giving the model the ability to say whether an image is recognizable or not.



Model: A cat laying on the floor next to a table

Reference: Quality issues are too severe to recognize visual content

Another problem we recognized was labels, the captions tend to name the exact brand of the object, and deduce what the contents are based on that brand.

For example our model confuses a hand sanitiser with a bottle of water, because it likely doesn't know what the brand germ-x is.



Model: A bottle of water sitting on top of a table

Reference: A bottle of germ-x sanitiser that is empty on the counter

2.2 Transformer Finetuning - Quantitative

As we've seen in the qualitative examples, the model describes scenes well, but most likely needs some adaptation. Because this dataset was taken by blind people, images are likely a lot different to the data it was trained on.

We first started by trying to fine tune the ViT + GP2 architecture we used on inference. We did some initial training tests with only the captioner unfrozen to land on some good hyperparameters as well as a good augmentation stack.

We landed on the following configuration.

```
lr: 5e-5
batch_size: 16
warmup_steps: 1000

augmentations:
- RandomHorizontalFlip(p=0.5)
- RandomAffine(degrees=5)
```

After landing on these hyperparameters, we did 3 long runs with early stopping, we tested fine tuning the captioner, the backbone, or all parameters.

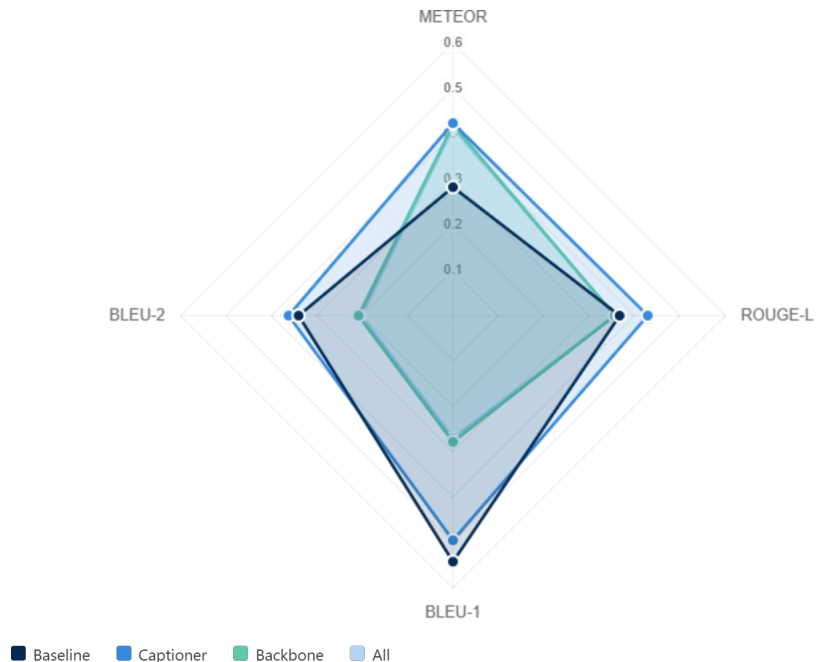
2.2 Transformer Finetuning - Quantitative

All three fine tuning strategies improved on the original model on the meteor benchmark, which we consider the most semantically important metric.

It is clear from the chart that the best fine tuning strategy came from fine tuning the captioner, because a part from the Bleu-1 metric, it beat the baseline on and both other methods.

This may be due to insufficient data to correctly train the backbone, or the fact that a lot of the images in this dataset are blurry and hard to make out.

Unfrozen	Meteor	Rouge-l	Bleu-1	Bleu-2
Baseline	0.283	0.367	0.542	0.340
Captioner	0.424	0.429	0.495	0.362
Backbone	0.420	0.356	0.278	0.208
All	0.414	0.357	0.270	0.202



2.2 Transformer Finetuning - Qualitative

After fine tuning the models, we wanted to see what had changed for the scores to become higher. We decided to check all three fine tuning strategies, to see how the different strategies adapted to this new dataset.



Captioner: Quality issues are to severe to recognize visual content

Backbone: A white wall with a brown and black object on it

All: A white wall with a black cord hanging from it

Reference: Quality issues are to severe to recognize visual content

It seems at least part of the improvement we see from the captioner fine tuning is the gained ability to point out images that are impossible to recognize

2.2 Transformer Finetuning - Qualitative



Captioner: Quality issues are to severe to recognize visual content

Backbone: A close up of a purple fabric with some wrinkles

All: A close up of a piece of fabric that is brown in color

Reference 1: Quality issues are to severe to recognize visual content

Reference 2: A blue piece of material sitting on some sort of surface

It also seems the captioner fine tuning may have caused the model to be more “afraid” of predicting, in this example training the entire model or the backbone make the model try to guess what is happening, and apart from the color mistake it is correct, but the fine tuned captioner doesn’t try to predict it.

2.3 Other Transformer Models

After testing and fine tuning the ViT+GPT2 model, we wanted to try some different captioning models. We tried paring different backbones with GPT2, changing GPT2 for other captioning modules, as well as the GIT model.

Unfortunately, none of this tests gave comparable results to the ViT+GPT2 model, changing the backbone or captioner makes the fine tuning process harder, as the two halves have to adapt to each other whilst learning the new dataset. As for GIT, we believe we may have made mistakes on implementation, as the results for a pretrained model shouldn't have been that low, but we didn't want to spend more time when our previous fine tune had already achieved good results.

With more testing and time, we believe they could have achieved similar if not better scores.

Model	Meteor	Rouge-l	Bleu-1	Bleu-2
ViT+GPT2	0.424	0.429	0.495	0.362
BEiT+GPT2	0.27	0.26	0.26	0.18
ViT+DistilGPT2	0.125	0.129	0.05	0.143
GIT	0.16	0.32	0.36	0.118

3. Task 2: Image Captioning with LLMs

3.1 Zero-shot Multimodal LLMs

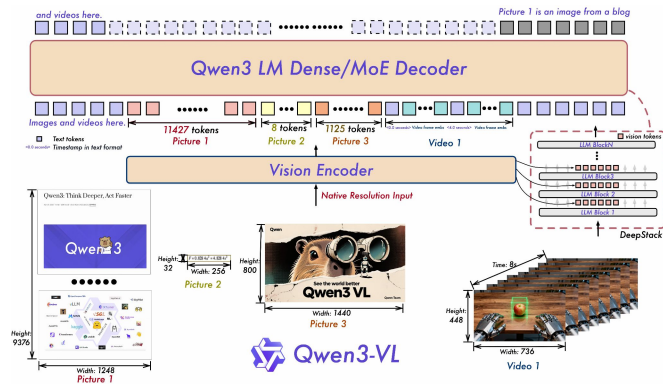
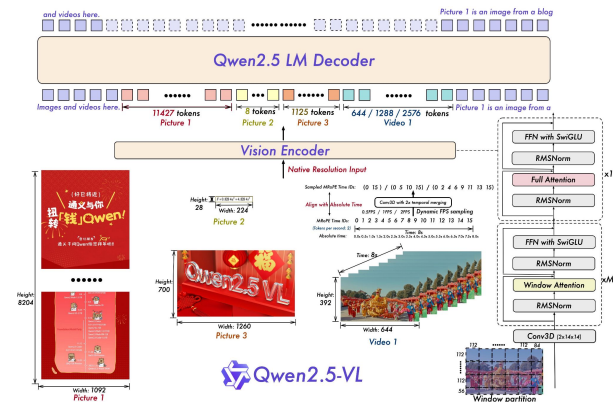
We evaluated three state-of-the-art Qwen vision-languages models in a zero-shot setting, without any fine-tuning on VizQiz.

Models evaluated:

- **Qwen2.5-VL-7B** (~14GB)
- **Qwen3-VL-8B** (~16GB)
- **Qwen3.5-9B** (~18GB)

To assess the impact of prompt engineering, we tested two strategies for each model:

- **Basic:** "Describe this image briefly"
- **Advanced:** domain-specific system prompt for VizWiz indicating that images may be blurry or poorly framed.

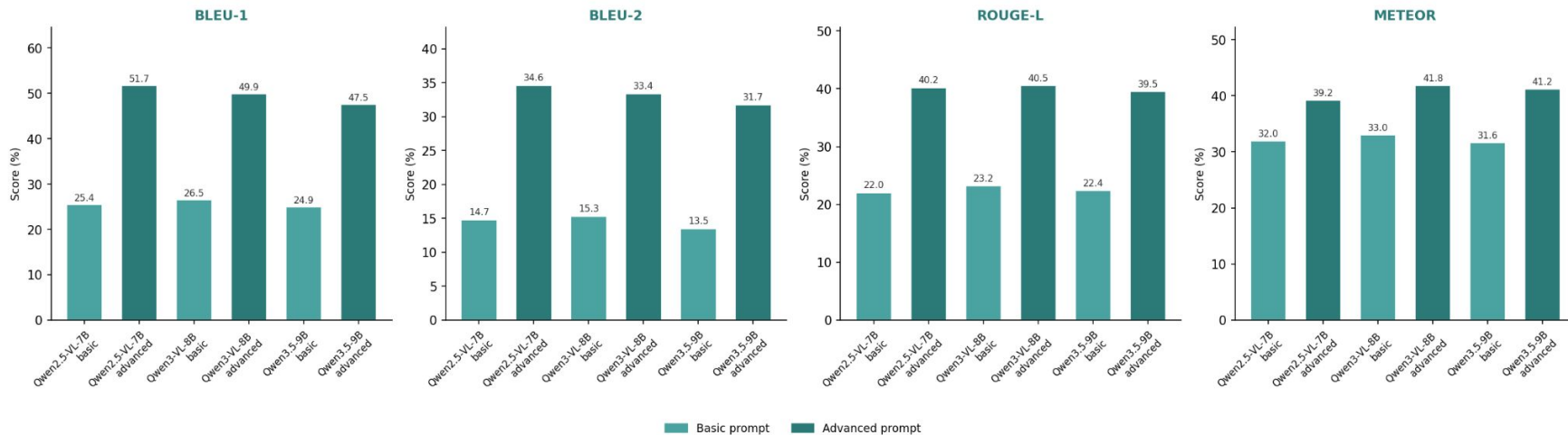


3.1 Zero-shot Multimodal LLMs

All three models achieve similar performance across all metrics, so we cannot identify a clear winner considering all metrics. However, we can see that all models benefit from using an advanced prompt instead of a basic one.

Focusing on **METEOR** (what we consider the most robust metric as it accounts for synonyms and paraphrasing), **Qwen3-VL-8B** achieves the best result (**41.8**), followed closely by **Qwen3.5-9B** (**41.2**) and **Qwen2.5-VL-7B** (**39.2**).

Task 2.1 — Direct evaluation: model & prompt comparison (VizWiz val)



3.1 Zero-shot Multimodal LLMs

We could see that providing domain-specific context in the prompt leads to important metric improvement across all models, with no additional computational cost.

The **advanced prompt** has two elements:

- **System prompt** that sets the context.
- **User instruction** that restricts the output format.

METEOR improves by **+7.2** to **+9.5** points depending on the model, which is a substantial gain for zero additional cost.

BLEU-1 and **BLEU-2** also improve a lot (BLEU-1 roughly doubles from **~25** to **~50**), which can be seen in the previous slide.

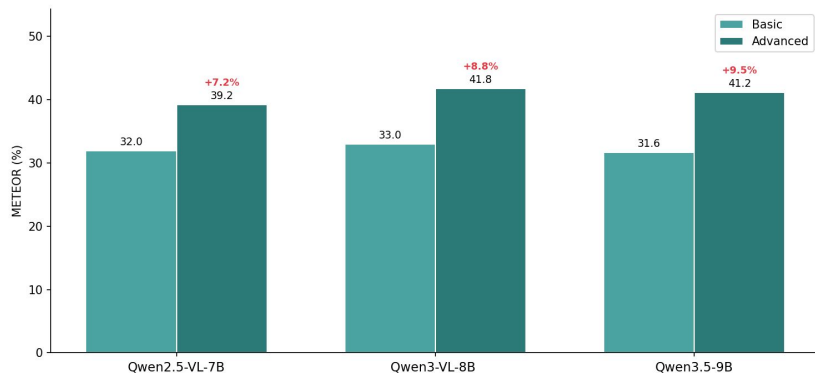
This suggests that large multimodal models already have strong visual understanding, and the bottleneck in zero-shot setting is often how the question is asked.

```
BASIC_PROMPT = Describe this image briefly.
```

```
ADVANCED_SYSTEM_PROMPT = You are an image captioning assistant helping visually impaired people understand their surroundings. Images may be blurry, poorly lit, or taken at unusual angles. Describe what you see concisely and factually, focusing on the main subject.
```

```
ADVANCED_USER_PROMPT = Describe this image in one short sentence. Output only the caption, nothing else.
```

Prompt engineering effect — basic vs advanced



3.1 Zero-shot Multimodal LLMs



References:

1. a plastic container of cookies with english and hebrew text
2. appears to be a picture of a container of cookies
3. a corner of a transparent plastic cookie box on a white countertop.

Caption Qwen3-VL-8B - Basic prompt

"this is a close-up photo of a plastic container of cookies, likely from israel, given the hebrew text on the label. the label features a blue and yellow design with clouds and the brand name "138" in red and white. it shows nutritional information per 100g"

Caption Qwen3-VL-8B - Advanced prompt

"a close-up of a hebrew-language cookie package showing nutritional information and ingredients."

The advanced prompt produces a concise and direct caption, very similar to the references style. The basic prompt, generates a verbose description, which is very different from the expected caption format.

3.1 Zero-shot Multimodal LLMs



References:

1. a bottle of germ-x hand sanitizer that is empty on the counter.
2. a clear plastic bottle containing germicide gel on a white computer desk with someone's name written in sharpie on the back
3. a bottle of hand sanitizer is almost empty

Qwen2.5-VL-7B - Advanced prompt

a bottle of germ-x hand sanitizer with a handwritten label is placed on a desk near a keyboard and mouse.

Qwen3-VL-8B - Advanced prompt

a clear bottle of germ-x hand sanitizer with "dapacom" handwritten on the label, sitting on a desk next to a keyboard.

Qwen3.5-9B - Advanced prompt

a bottle of germ-x hand sanitizer with a blue cap, sitting on a desk next to a keyboard and other office items.

All three models produce accurate captions similar to the references, correctly identifying the brand "Germ-X" and the desk context. Qwen3-VL-8B also detects the handwritten name on the label, an extra detail that helps explain its slightly higher METEOR score.

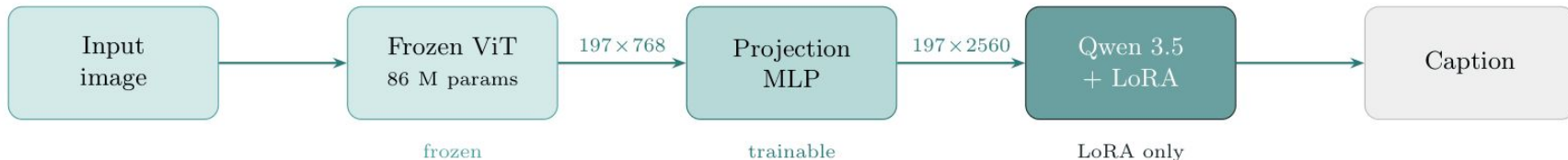
3.2 ViT + LoRA-adapted Qwen3.5

The architecture combines a frozen **ViT encoder** (*nlpconnect/vit-gpt2-image-captioning*, 86M params) with a **Qwen3.5** text decoder connected via a learned **Projection MLP**.

Since the best **Task 1.2** result was only fine-tuning the **GPT2** decoder, the ViT weights were never updated, so we reuse them directly without any additional training.

The Projection MLP (2 linear layers, trained from scratch) maps the ViT output space (197×768) to the Qwen embedding space. This enabled us to test 2 different decoder sizes: **Qwen3.5-0.8B** (~1.5GB) and **Qwen3.5-4B** (~8GB).

Training was performed on a NVIDIA H100 (80GB) with batch size 8, consuming over 50GB of VRAM. Each epoch took approximately 1 hour, with an additional 30 minutes for validation, so we decided to limit our experiment to 5 epochs per configuration.



3.2 ViT + LoRA-adapted Qwen3.5

We explored different combinations of **rank r** and **scaling factor α** for both decoder sizes, keeping $lr=1e-4$ and $dropout=0.05$. A lower rank means fewer trainable parameters but less capacity, while a higher α increases the effective learning rate of the LoRA adapters.

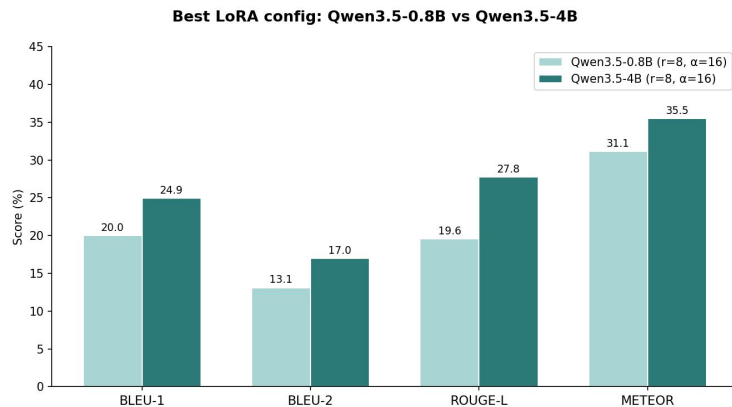
The best configuration for both models was **$r=8$, $\alpha=16$** . We think lower rank generalizes better on the relatively small VizWiz training set, suggesting that higher capacity leads to overfitting rather than improvement.

Qwen3.5-0.8B	r	α	Meteor	Rouge-l	Bleu-1	Bleu-2
	8	16	0.311	0.196	0.200	0.131
	8	32	0.298	0.209	0.204	0.129
	16	16	0.255	0.179	0.208	0.120
	16	32	0.289	0.189	0.195	0.124
Qwen3.5-4B	r	α	Meteor	Rouge-l	Bleu-1	Bleu-2
	8	16	0.355	0.278	0.249	0.170
	8	32	0.338	0.259	0.241	0.157
	16	16	0.304	0.247	0.218	0.147
	16	32	0.334	0.272	0.239	0.158

3.2 ViT + LoRA-adapted Qwen3.5

Comparing the best LoRA configuration ($r=8$, $\alpha=16$) across both decoder sizes, **Qwen3.5-4B** outperforms **Qwen3.5-0.8B** on all metrics, with a **METEOR** improvement of **+4.4** points (35.5 vs 31.1).

Given this consistent advantage, we select **Qwen3.5-4B** as our decoder for all the subsequent experiments on learning rate and dropout.

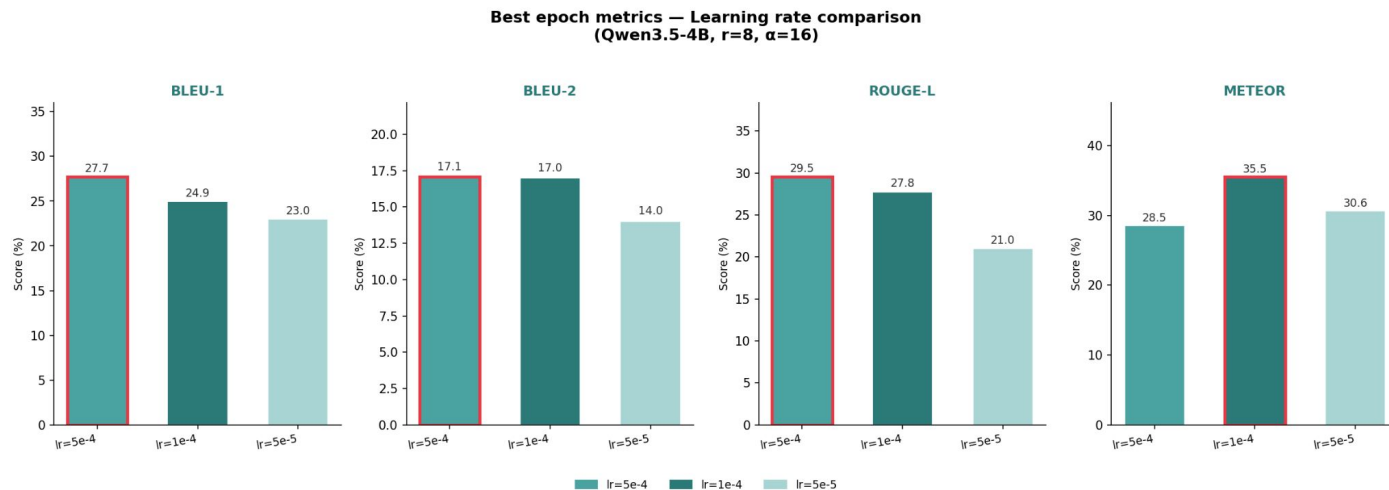


3.2 ViT + LoRA-adapted Qwen3.5

Using the best configuration found in the previous experiment (Qwen3.5-4B, $r=8$, $\alpha=16$), we tested three **learning rates**: 5e-4, 1e-4 and 5e-5.

With $lr=5e-4$ we achieved the highest scores on BLEU-1 (27.7), BLEU-2 (17.1) and ROUGE-L (29.5). However, focusing on **METEOR**, $lr=1e-4$ clearly wins with 35.5, outperforming $lr=5e-4$ (28.5) by 7 points and $lr=5e-5$ (30.6) by nearly 5 points.

This confirms $lr=1e-4$ as the **optimal learning rate** for all subsequent experiments.

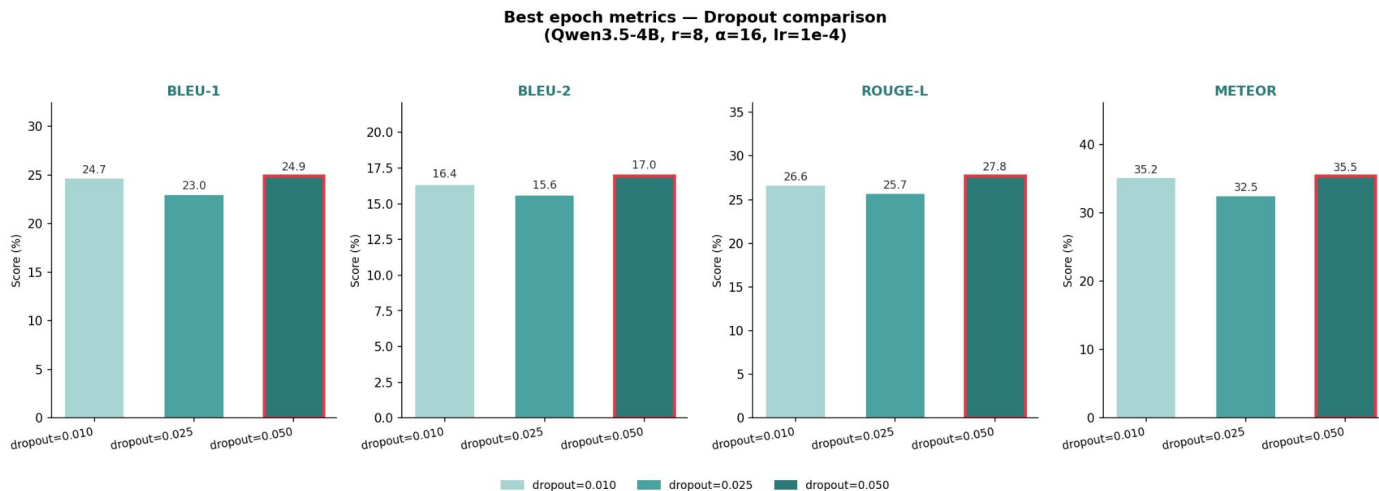


3.2 ViT + LoRA-adapted Qwen3.5

Using the optimal learning rate found in the previous experiment, we evaluated three **dropout values**: 0.010, 0.025 and 0.050.

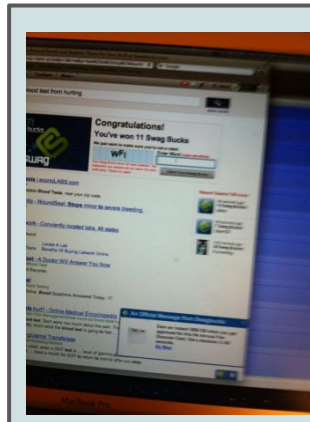
The results are very similar across all three values, with differences of less than 3 points on any metric. Dropout of **0.050** achieves the **best scores on all four metrics**, even though it is just by a small margin over dropout=0.010 and dropout=0.025.

This small difference suggests that dropout has a limited impact in this configuration, the model does not seem to be overfitting within 5 epochs. With more training epochs we would likely see a clearer difference, as a higher dropout would help regularize a converged model.



3.2 ViT + LoRA-adapted Qwen3.5

Best model: Qwen3.5-4B, $r=8$, $\alpha=16$, $lr=1e-4$, dropout=0.05



References:

1. someone was doing a google and they received a message from swagbucks saying they won.
2. a picture of computer screen that has swag bucks on display.
3. a computer monitor screen showing " congratulations! you've won 11 swag bucks"

Prediction:

"assistant

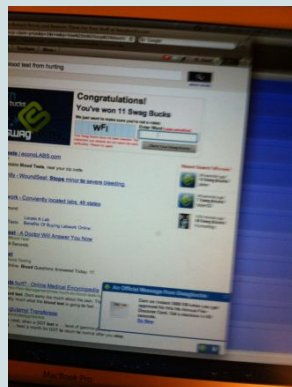
a computer screen with a website that is asking for a user's name and email address. the website is for a company called "the online store". the website is in a language that is not english. the website is on a computer screen that is blue and white. the website is on a computer"

Despite the artifacts, the model shows some understanding of the image, identifying a computer screen with a website. However, two clear generation issues are visible:

- **Chat template artifacts:** The output starts with "assitant", which is a token from Qwen3.5's chat template.
- **Verbosity and loops:** The model repeats the sam sentence structure multiple times.

3.2 ViT + LoRA-adapted Qwen3.5

Best model: Qwen3.5-4B, $r=8$, $\alpha=16$, $lr=1e-4$, dropout=0.05 (with post-processing)



References:

1. someone was doing a google and they received a message from swagbucks saying they won.
2. a picture of computer screen that has swag bucks on display.
3. a computer monitor screen showing " congratulations! you've won 11 swag bucks"

Prediction:

"a computer screen with a website that is asking for an email address."

Prediction with best model Task 2.1:

"a computer screen displaying a swagbucks website claiming the user won 11 swag bucks, with a captcha verification prompt."

Applying a **repetition penalty** and **removing chat template artifacts** produces a much cleaner caption, closer to the references. The output is concise, and demonstrates that the model has learned a useful visual-text mapping during fine-tuning.

However, it remains **less precise** than the **zero-shot results** from **Task 2.1**, where the model was able to capture the detail of the images. This suggests that while LoRA fine-tuning teaches the model the caption format, it does not completely compensate for the richer visual understanding of a trained multimodal model.

3.2 ViT + LoRA-adapted Qwen3.5

Best model: Qwen3.5-4B, $r=8$, $\alpha=16$, $lr=1e-4$, dropout=0.05

	Meteor	Rouge-l	Bleu-1	Bleu-2
Without post-processing	0.355	0.278	0.249	0.170
With post-processing	0.382	0.441	0.572	0.423

Post-processing produces a big improvement on **BLEU** and **ROUGE-L** metrics, which are very sensitive to caption length. **METEOR** improves just a bit, as it already takes into account paraphrasing and is less affected by verbosity.

Ideally, post-processing should have been applied across all hyperparameter search experiments. However, due to time constraints we could not re-run all configurations. It is important to remark that all our hyperparameter decisions were based on METEOR, which is the least affected metric by verbosity, with only a small difference between the two configurations.

The ranking between experiments remains valid since all were evaluated under the same conditions, so the conclusions hold regardless of the absolute metric values.

4. Comparison and Conclusions

General Discuss and Recap

To recap our results, we noticed a clear hierarchy between the different approaches we tested. In the first task with the ViT-GPT2 architecture, fine-tuning simply the captioner gave us the most robust results and allowed the model to successfully detect too blurry images, while training the backbone actually caused a big loss in performance.

We also saw that trying other pairs of encoder-decoders wasn't profitable, as forcing two different halves to adapt to each other on a new dataset makes the fine-tuning process much harder.

Moving to the MLLMs in Task 2, zero-shot evaluation showed that using an advanced prompt is strictly necessary to force the model to be brief and avoid over-complicated descriptions.

Finally, for our custom PEFT pipeline, we found that a smaller rank of 8 and alpha of 16 generalized better for Qwen3.5-4B and tested the importance of a good postprocessing in LLMs, and most importantly proving the efficiency of LoRA for these purposes, although the training time was still very high.

Task	Architecture	Meteor	Rouge-L	Bleu-1	Bleu-2
Task 1	Baseline	0.283	0.367	0.542	0.340
	Captioner	0.424	0.429	0.495	0.362
	Backbone	0.420	0.356	0.278	0.208
	All	0.414	0.357	0.270	0.202
Task 2	Zero Shot	0.415	0.405	0.499	0.334
	LoRA	0.382	0.441	0.572	0.423

Summary Slide



Conclusions

ViT+GPT2 Inference

- Decent initial metrics
- Problems with labeled items like food or bottles

ViT+GPT2 Training

- Finetune only the captioner
- Learnt some images are too messy

Zero Shot MLLMs

- Important to give a good prompt
- Tendency to over describe

LoRA MLLMs

- Enables adapting large models
- Training remains time consuming



Qualitative:



Best model Task 1 (ViT & GPT2):

"a bottle of water sitting on a desk next to a computer keyboard and mouse"

Best model Task 2 (Qwen3.5-9B - Advanced prompt):

"a clear bottle of germ-x hand sanitizer with 'dapacom' handwritten on the label, sitting on a desk next to a keyboard."



Quantitative :

